

D1 Binomial expansion

Name: _____

Class: _____

Date: _____

Time: **324 minutes**

Marks: **274 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) Find the first three terms in the expansion of $(1 - 3x)^4$ in ascending powers of x .

(3)

- (b) Using your expansion, approximate $(0.994)^4$ to six decimal places.

(2)

(Total 5 marks)

Q2.

In the binomial expansion of $(1 + x)^n$, where $n \geq 4$, the coefficient of x^4 is $1\frac{1}{2}$ times the sum of the coefficients of x^2 and x^3

Find the value of n .

(Total 5 marks)

Q3.

- (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$\frac{1}{\sqrt{4+x}}$$

(3)

- (b) Hence, find the first three terms of the binomial expansion of $\frac{1}{\sqrt{4-x^3}}$

(2)

- (c) Using your answer to part (b), find an approximation for $\int_0^1 \frac{1}{\sqrt{4-x^3}} dx$, giving your answer to seven decimal places.

(3)

- (d) (i) Edward, a student, decides to use this method to find a more accurate value for the integral by increasing the number of terms of the binomial expansion used.

Explain clearly whether Edward's approximation will be an overestimate, an underestimate, or if it is impossible to tell.

(2)

- (ii) Edward goes on to use the expansion from part (b) to find an approximation

for $\int_{-2}^0 \frac{1}{\sqrt{4-x^3}} dx$

Explain why Edward's approximation is invalid.

(2)

(Total 12 marks)

Q4.

Find the coefficient of x^2 in the expansion of $(1 + 2x)^7$

Circle your answer.

42

4

21

84

(Total 1 mark)

Q5.

(a) Find the first **three** terms, in ascending powers of x , of the expansion of $(1 - 2x)^{10}$

(3)

(b) Carly has lost her calculator. She uses the first three terms, in ascending powers of x , of the expansion of $(1 - 2x)^{10}$ to evaluate 0.998^{10}
Find Carly's value for 0.998^{10} and show that it is correct to **five** decimal places.

(3)

(Total 6 marks)

Q6.

State the values of $|x|$ for which the binomial expansion of $(3 + 2x)^{-4}$ is valid.
Circle your answer.

$|x| < \frac{2}{3}$

$|x| < 1$

$|x| < \frac{3}{2}$

$|x| < 3$

(Total 1 mark)

Q7.

(a) Find the first three terms, in ascending powers of x , in the binomial expansion of $(1 + 6x)^{\frac{1}{3}}$

(2)

(b) Use the result from part (a) to obtain an approximation to $\sqrt[3]{1.18}$ giving your answer to 4 decimal places.

(2)

(c) Explain why substituting $x = \frac{1}{2}$ into your answer to part (a) does not lead to a valid approximation for $\sqrt[3]{4}$

(1)

(Total 5 marks)

Q8.

- (a) Expand $\left(1 + \frac{4}{x}\right)^2$. (1)
- (b) The first four terms of the binomial expansion of $\left(1 + \frac{x}{4}\right)^8$ in ascending powers of x are $1 + ax + bx^2 + cx^3$. Find the values of the constants a , b and c . (4)
- (c) Hence find the coefficient of x in the expansion of $\left(1 + \frac{4}{x}\right)^2 \left(1 + \frac{x}{4}\right)^8$. (4)
- (Total 9 marks)

Q9.

It is given that $f(x) = \frac{7x - 1}{(1 + 3x)(3 - x)}$

- (a) Express $f(x)$ in the form $\frac{A}{3 - x} + \frac{B}{1 + 3x}$, where A and B are integers. (3)
- (b) (i) Find the first three terms of the binomial expansion of $f(x)$ in the form $a + bx + cx^2$, where a , b and c are rational numbers. (7)
- (ii) State why the binomial expansion cannot be expected to give a good approximation to $f(x)$ at $x = 0.4$. (1)
- (Total 11 marks)

Q10.

- (a) (i) Using the binomial expansion, or otherwise, express $(2 + y)^3$ in the form $a + by + cy^2 + y^3$, where a , b and c are integers. (2)
- (ii) Hence show that $(2 + x^{-2})^3 + (2 - x^{-2})^3$ can be expressed in the form $p + qx^{-4}$, where p and q are integers. (3)
- (b) (i) Hence find $\int \left[(2 + x^{-2})^3 + (2 - x^{-2})^3 \right] dx$. (2)

- (ii) Hence find the value of $\int_1^2 [(2+x^{-2})^3 + (2-x^{-2})^3] dx$.
- (2)
(Total 9 marks)

Q11.

- (a) Find the binomial expansion of $(1+6x)^{-\frac{1}{3}}$ up to and including the term in x^2 .
(2)
- (b) (i) Find the binomial expansion of $(27+6x)^{-\frac{1}{3}}$ up to and including the term in x^2 , simplifying the coefficients.
(3)
- (ii) Given that $\sqrt[3]{\frac{2}{7}} = \frac{2}{\sqrt[3]{28}}$, use your binomial expansion from part (b)(i) to obtain an approximation to $\sqrt[3]{\frac{2}{7}}$, giving your answer to six decimal places.
(2)
(Total 7 marks)

Q12.

- (a) (i) Describe the geometrical transformation that maps the graph of $y = \left(1 + \frac{x}{3}\right)^6$ onto the graph of $y = (1+2x)^6$.
(2)
- (ii) The curve $y = \left(1 + \frac{x}{3}\right)^6$ is translated by the vector $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ to give the curve $y = g(x)$.
Find an expression for $g(x)$, simplifying your answer.
(2)
- (b) The first four terms in the binomial expansion of $\left(1 + \frac{x}{3}\right)^6$ are $1 + ax + bx^2 + cx^3$.
Find the values of the constants a , b and c , giving your answers in their simplest form.
(4)
(Total 8 marks)

Q13.

- (a) Find the binomial expansion of $(1 + 6x)^{\frac{2}{3}}$ up to and including the term in x^2 . (2)
- (b) Find the binomial expansion of $(8 + 6x)^{\frac{2}{3}}$ up to and including the term in x^2 . (3)
- (c) Use your answer from part (b) to find an estimate for $\sqrt[3]{100}$ in the form $\frac{a}{b}$, where a and b are integers. (2)
- (Total 7 marks)**

Q14.

- (a) Expand $\left(x^{\frac{3}{2}} - 1\right)^2$. (2)
- (b) Hence find $\int \left(x^{\frac{3}{2}} - 1\right)^2 dx$. (3)
- (c) Hence find the value of $\int_1^4 \left(x^{\frac{3}{2}} - 1\right)^2 dx$. (2)
- (Total 7 marks)**

Q15.

- (a) Find the binomial expansion of $(1 + 4x)^{\frac{1}{2}}$ up to and including the term in x^2 . (2)
- (b) (i) Find the binomial expansion of $(4 - x)^{-\frac{1}{2}}$ up to and including the term in x^2 . (3)
- (ii) State the range of values of x for which the expansion in part (b)(i) is valid. (1)
- (c) Find the binomial expansion of $\sqrt{\frac{1 + 4x}{4 - x}}$ up to and including the term in x^2 . (2)
- (Total 8 marks)**

Q16.

- (a) Using the binomial expansion, or otherwise, express $(1 - x)^3$ in ascending powers of x . (2)
- (b) Show that the expansion of $(1 + y)^4 - (1 - y)^3$ is $7y + py^2 + qy^3 + y^4$ where p and q are constants to be found. (4)
- (c) Hence find $\int \left[(1 + \sqrt{x})^4 - (1 - \sqrt{x})^3 \right] dx$, expressing each coefficient in its simplest form. (4)
- (Total 10 marks)**

Q17.

- (a) Express $\frac{3 + 9x}{(1 + x)(3 + 5x)}$ in the form $\frac{A}{1 + x} + \frac{B}{3 + 5x}$, where A and B are integers. (3)
- (b) Hence, or otherwise, find the binomial expansion of $\frac{3 + 9x}{(1 + x)(3 + 5x)}$ up to and including the term in x^2 . (7)
- (c) Find the range of values of x for which the binomial expansion of $\frac{3 + 9x}{(1 + x)(3 + 5x)}$ is valid. (2)
- (Total 12 marks)**

Q18.

- (a) The expression $(2 + x^2)^3$ can be written in the form $8 + px^2 + qx^4 + x^6$. Show that $p = 12$ and find the value of the integer q . (3)

- (b) (i) Hence find $\int \frac{(2+x^2)^3}{x^4} dx$. (5)

- (ii) Hence find the exact value of $\int_1^2 \frac{(2+x^2)^3}{x^4} dx$. (2)
(Total 10 marks)

Q19.

- (a) (i) Find the binomial expansion of $(1-x)^{\frac{1}{3}}$ up to and including the term in x^2 . (2)
- (ii) Hence, or otherwise, show that

$$(125 - 27x)^{\frac{1}{3}} \approx 5 + \frac{m}{25}x + \frac{n}{3125}x^2$$

for small values of x , stating the values of the integers m and n . (3)

- (b) Use your result from part (a)(ii) to find an approximate value of $\sqrt[3]{119}$, giving your answer to five decimal places. (2)
(Total 7 marks)

Q20.

- (a) Expand $(5+h)^3$. (1)

- (b) A curve has equation $y = x^3 - x^2$.
- (i) Find the gradient of the line passing through the point (5, 100) and the point on the curve for which $x = 5 + h$. Give your answer in the form $p + qh + rh^2$ where p , q and r are integers. (4)

- (ii) Show how the answer to part (b)(i) can be used to find the gradient of the curve at the point (5, 100). State the value of this gradient. (2)
(Total 7 marks)

Q21.

- (a) The first four terms of the binomial expansion of $(1 + 2x)^7$ in ascending powers of x are $1 + ax + bx^2 + cx^3$. Find the values of the integers a , b and c .

(4)

- (b) Hence find the coefficient of x^3 in the expansion of $\left(1 - \frac{1}{2}x\right)^2 (1 + 2x)^7$.

(4)

(Total 8 marks)

Q22.

- (a) (i) Find the binomial expansion of $(1 + x)^{-\frac{1}{3}}$ up to and including the term in x^2 .

(2)

- (ii) Hence find the binomial expansion of $\left(1 + \frac{3}{4}x\right)^{-\frac{1}{3}}$ up to and including the term in x^2 .

(2)

- (b) Hence show that $\sqrt[3]{\frac{256}{4 + 3x}} \approx a + bx + cx^2$ for small values of x , stating the values of the constants a , b and c .

(3)

(Total 7 marks)

Q23.

- (a) The expression $\left(1 - \frac{1}{x^2}\right)^3$ can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} - \frac{1}{x^6}$$

Find the values of the integers p and q .

(2)

- (b) (i) Hence find $\int \left(1 - \frac{1}{x^2}\right)^3 dx$.

(4)

- (ii) Hence find the value of $\int_{\frac{1}{2}}^1 \left(1 - \frac{1}{x^2}\right)^3 dx$.

(2)

(Total 8 marks)

Q24.

- (a) (i) Find the binomial expansion of $(1+x)^{\frac{3}{2}}$ up to and including the term in x^2 . (2)

- (ii) Find the binomial expansion of $(16+9x)^{\frac{3}{2}}$ up to and including the term in x^2 . (3)

- (b) Use your answer to part (a)(ii) to show that $13^{\frac{3}{2}} \approx 46 + \frac{a}{b}$, stating the values of the integers a and b . (2)

(Total 7 marks)

Q25.

- (a) By using the binomial expansion, or otherwise, express $(1+2x)^4$ in the form

$$1 + ax + bx^2 + cx^3 + 16x^4$$

where a , b and c are integers. (4)

- (b) Hence show that $(1+2x)^4 + (1-2x)^4 = 2 + 48x^2 + 32x^4$. (3)

- (c) Hence show that the curve with equation

$$y = (1+2x)^4 + (1-2x)^4$$

has just one stationary point and state its coordinates. (4)

(Total 11 marks)

Q26.

- (a) (i) Find the binomial expansion of $(1-x)^{\frac{1}{2}}$ up to and including the term in x^2 . (2)

- (ii) Hence obtain the binomial expansion of $\sqrt{4-x}$ up to and including the term in x^2 . (3)

- (b) Use your answer to part (a)(ii) to find an approximate value for $\sqrt[3]{3}$. Give your answer to three decimal places.

(2)

(Total 7 marks)

Q27.

- (a) Write down the value of n given that $\frac{1}{x^4} = x^n$.

(1)

- (b) Expand $\left(1 + \frac{3}{x^2}\right)^2$.

(2)

- (c) Hence find $\int \left(1 + \frac{3}{x^2}\right)^2 dx$.

(3)

- (d) Hence find the exact value of $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx$.

(2)

(Total 8 marks)

Q28.

- (a) Find the binomial expansion of $(1 - x)^{-1}$ up to and including the term in x^2 .

(2)

- (b) (i) Express $\frac{3x-1}{(1-x)(2-3x)}$ in the form $\frac{A}{1-x} + \frac{B}{2-3x}$, where A and B are integers.

(3)

- (ii) Find the binomial expansion of $\frac{3x-1}{(1-x)(2-3x)}$ up to and including the term in x^2 .

(6)

- (c) Find the range of values of x for which the binomial expansion of $\frac{3x-1}{(1-x)(2-3x)}$ is valid.

(2)

(Total 13 marks)

Q29.

(a) Using the binomial expansion, or otherwise:

(i) express $(1 + x)^3$ in ascending powers of x ;

(2)

(ii) express $(1 + x)^4$ in ascending powers of x .

(2)

(b) Hence, or otherwise:

(i) express $(1 + 4x)^3$ in ascending powers of x ;

(2)

(ii) express $(1 + 3x)^4$ in ascending powers of x .

(2)

(c) Show that the expansion of

$$(1 + 3x)^4 - (1 + 4x)^3$$

can be written in the form

$$px^2 + qx^3 + rx^4$$

where p , q and r are integers.

(2)

(Total 10 marks)

Q30.

(a) Obtain the binomial expansion of $(1 + x)^{\frac{1}{2}}$ up to and including the term in x^2 .

(2)

(b) Hence obtain the binomial expansion of $\sqrt{1 + \frac{3}{2}x}$ up to and including the term in x^2 .

(2)

(c) Hence show that $\sqrt{\frac{2 + 3x}{8}} \approx a + bx + cx^2$ for small values of x , where a , b and c are constants to be found.

(2)

(Total 6 marks)

Q31.

- (a) The expression $\left(1 + \frac{4}{x^2}\right)^3$ can be written in the form

$$1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^6}$$

By using the binomial expansion, or otherwise, find the values of the integers p and q .

(3)

- (b) (i) Hence find $\int \left(1 + \frac{4}{x^2}\right)^3 dx$.

(4)

- (ii) Hence find the value of $\int_1^2 \left(1 + \frac{4}{x^2}\right)^3 dx$.

(2)

(Total 9 marks)

Q32.

- (a) (i) Obtain the binomial expansion of $(1 - x)^{\frac{1}{4}}$ up to and including the term in x^2 .

(2)

- (ii) Hence show that $(81 - 16x)^{\frac{1}{4}} \approx 3 - \frac{4}{27}x - \frac{8}{729}x^2$ for small values of x .

(3)

- (b) Use the result from part (a)(ii) to find an approximation for $\sqrt[4]{80}$, giving your answer to seven decimal places.

(2)

(Total 7 marks)

Q33.

- (a) The first four terms of the binomial expansion of $(1 + 2x)^8$ in ascending powers of x are $1 + ax + bx^2 + cx^3$. Find the values of the integers a , b and c .

(4)

- (b) Hence find the coefficient of x^3 in the expansion of $\left(1 + \frac{1}{2}x\right)(1 + 2x)^8$.

(3)

(Total 7 marks)

Q34.

- (a) Find the binomial expansion of $(1+x)^{\frac{1}{3}}$ up to the term in x^2 . (2)

- (b) (i) Show that $(8+3x)^{\frac{1}{3}} \approx 2 + \frac{1}{4}x - \frac{1}{32}x^2$ for small values of x . (3)

- (ii) Hence show that $\sqrt[3]{9} \approx \frac{599}{288}$. (2)

(Total 7 marks)

Q35.

A curve is defined for $x > 0$ by the equation

$$y = \left(1 + \frac{2}{x}\right)^2$$

The point P lies on the curve where $x = 2$.

- (a) Find the y -coordinate of P . (1)

- (b) Expand $\left(1 + \frac{2}{x}\right)^2$. (2)

- (c) Find $\frac{dy}{dx}$. (3)

- (d) Hence show that the gradient of the curve at P is -2 . (2)

- (e) Find the equation of the normal to the curve at P , giving your answer in the form $x + by + c = 0$, where b and c are integers. (4)

(Total 12 marks)